

Spain Energy Grid Risk Index: Next-Day Forecasting System

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Context & Problem

Operators lack a unified measure of **grid stress**

Continuity

Operators must balance generation and demand variability and ensure continuity

Limited prediction

Current forecasts are limited to energy demand and generation, **ignoring stress on the network**

Risk

Reactive management, limited planning capacity and exposure to failures in the network

External factors

Seasonal, meteorological and economic variables influence tension in the network

Our proposal

A **Grid Risk Index** to measure grid tension

Flexible share

The proportion of total generation from dispatchable and adaptable sources

+

Forecast Error

Quantifies deviation between expected and actual demand

+

Net Load

Captures structural pressure on dispatchable generation

Weather

Energy price

Lagged features

Model

Building a predictive model for the GRI

Features

Target: GRI

Net Load

Forecast error

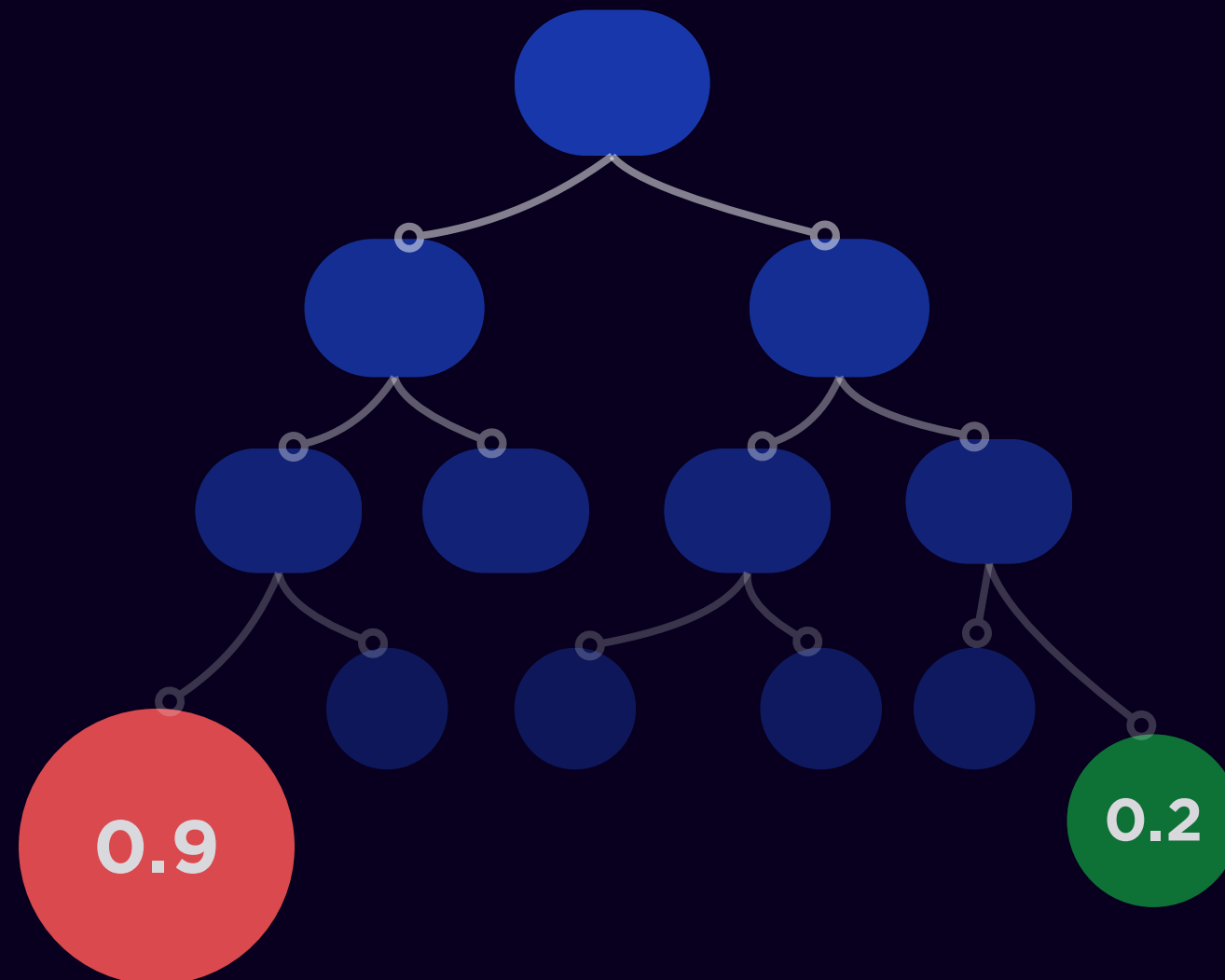
Flexible share

+

Seasonal, weather and econometric features

Model

Gradient Boosting Machine



Prediction

Classify GRI

Low ($<20p$)

Stable (40p)

Elevated (60p)

Severe (80p)

Extreme (>80p)

Business value

From prediction to operational decision support

Comprehensive

Red Eléctrica counts with a prediction product of a wide set of features, providing a robust measure of grid tension.

Reliable

The pipeline integrates the latest data available from official sources, offering a reliable day-ahead prediction.

Interpretable

The index is designed to be easily interpretable, facilitating proactive grid management and infrastructure planning.

Technical walkthrough

Building an end-to-end pipeline for our model

